

MCS-230

Digital Image Processing & Computer Vision

Comprehensive PYQ Solutions Book

Based on Frequency Analysis of 7 Semesters (2022 – 2025)

MCA (New) Programme | IGNOU

Semesters Covered: Dec 2022 | Jun 2023 | Dec 2023 | Jun 2024 | Dec 2024 | Jun 2025 | Dec 2025

Topic Frequency Analysis — 7 Semesters

The table below ranks every topic by how many times it appeared across all 7 question papers. Topics marked ★★★ are exam-critical and answered in full detail in later chapters.

Rank	Topic	Total	D22	J23	D23	J24	D24	J25	D25	Priority
1	Image Digitization (Sampling & Quantization)	7/7	✓	✓	✓	✓	✓	✓	✓	★★★
2	DFT (Discrete Fourier Transform)	7/7	✓	✓	✓	✓	✓	✓	✓	★★★
3	Image Enhancement (Spatial Domain)	7/7	✓	✓	✓	✓	✓	✓	✓	★★★
4	Noise Models (Gaussian, Salt & Pepper, etc.)	7/7	✓	✓	✓	✓	✓	✓	✓	★★★
5	K-Means Clustering	6/7	✓	✓	✓	✓	✓	✓	–	★★★
6	Supervised vs Unsupervised Learning	6/7	✓	✓	✓	✓	–	✓	✓	★★★
7	Colour Models (RGB, CMY, HSI, CMYK)	6/7	✓	–	✓	✓	✓	✓	✓	★★★
8	Laplacian Filter / Image Sharpening	5/7	–	–	–	–	✓	–	✓	★★★
9	Histogram Equalization / Matching	5/7	–	–	–	✓	✓	–	–	★★★
10	Wavelet Transform (DWT/CWT)	5/7	–	✓	✓	–	✓	✓	–	★★★
11	Agglomerative Hierarchical Clustering	5/7	✓	–	✓	–	–	–	✓	★★★
12	Frequency Domain Filters (LPF/HPF)	5/7	–	✓	✓	✓	✓	–	✓	★★
13	Image Degradation & Restoration	4/7	✓	✓	✓	✓	–	–	–	★★
14	Camera Models (Pin-hole / Calibration)	4/7	–	–	–	–	–	✓	✓	★★
15	Edge Detection / Segmentation	4/7	–	–	–	–	–	✓	✓	★★
16	Haar Transform	3/7	✓	–	–	–	–	✓	–	★★
17	DCT (Discrete Cosine Transform)	3/7	–	✓	–	✓	✓	–	–	★★
18	Image Acquisition Systems	3/7	–	✓	✓	–	–	–	✓	★★
19	Linear Discriminant Function	3/7	✓	✓	✓	–	–	–	–	★
20	Wiener Filter	2/7	–	–	–	–	–	–	✓	★
21	Homography	2/7	–	–	–	–	–	–	✓	★
22	Bayesian Classifier	2/7	–	✓	–	✓	–	–	–	★
23	Feature Extraction & Object Detection	2/7	–	–	–	–	–	✓	–	★
24	Stereovision / Depth Perception	2/7	–	–	–	–	–	✓	–	★

UNIT 1: Image Acquisition & Digitization

Appeared in: ALL 7 Semesters | Priority: ★★★

1.1 Image Acquisition

Image acquisition is the process of capturing an image from the real world using an imaging sensor and converting it into a digital form suitable for processing.

Types of Imaging Systems

Type	Description
Direct Imaging	The image sensor is placed directly in contact with or in close proximity to the object. Example: Medical X-ray films, flatbed scanners. Light from the source passes directly through or reflects off the object onto the sensor.
Indirect Imaging	Uses intermediate steps to capture the image. Example: CCD cameras, telescope imaging. Light is focused using lenses/mirrors before hitting the sensor.
Optical Imaging	Uses visible light. Example: Cameras, microscopes. Operates on the principle of reflection/refraction.
Analog Imaging	Produces continuous-signal output (not discrete). Example: Old CRT cameras. Requires ADC for digitisation.
Digital Imaging	Directly produces discrete (digital) output. Example: DSLR cameras, smartphones. Values stored as binary pixels.

1.2 Image Digitization — Sampling & Quantization

Digitization converts a continuous analog image into a discrete digital image through two steps:

Sampling

Sampling is the process of measuring the image intensity at discrete spatial locations. It determines the spatial resolution (number of pixels).

If an image has M rows and N columns:

$$\text{Total samples} = M \times N$$

Shannon-Nyquist Theorem:

$$\text{Sampling rate} \geq 2 \times (\text{maximum frequency in the signal})$$

$$f_s \geq 2 \times f_{\max}$$

Quantization

Quantization assigns a discrete integer value (grey level) to each sampled point. It determines the intensity resolution (number of grey levels).

Number of grey levels = 2^k (where k = number of bits)

For 8-bit image: grey levels = $2^8 = 256$ (0 to 255)

For 1-bit image: grey levels = $2^1 = 2$ (binary image)

Storage size = $M \times N \times k$ bits

Solved Numerical — Image Size Calculation

Problem (Dec 2024 / Jun 2023): Image is 5×8 inches, frequency = 500 dpi

Solution:

Samples in X direction = $5 \times 500 = 2500$

Samples in Y direction = $8 \times 500 = 4000$

Total samples = $2500 \times 4000 = 10,000,000 = 10$ million

Problem (Jun 2024 / Dec 2023): Aspect ratio 6:2, pixel resolution = 480,000

Solution:

Let width = $6k$ and height = $2k$

$6k \times 2k = 480,000$

$12k^2 = 480,000 \Rightarrow k^2 = 40,000 \Rightarrow k = 200$

Width = $6 \times 200 = 1200$ pixels

Height = $2 \times 200 = 400$ pixels

Image dimensions: 1200×400 pixels

Image size (8-bit) = $1200 \times 400 \times 8 = 3,840,000$ bits = 480,000 bytes ≈ 469 KB

Problem (Dec 2024): Smartphone 1080×1920 px, 5.0 inch diagonal — find PPI

Solution:

Diagonal in pixels = $\sqrt{1080^2 + 1920^2}$

$= \sqrt{1,166,400 + 3,686,400}$

$= \sqrt{4,852,800} \approx 2203$ pixels

PPI = $2203 / 5.0 = 440.6$ PPI

1.3 Types of Images

Type	Bits/Pixel	Grey Levels	Description
Binary Image	1	2	Black and white only. Used in document scanning.

Greyscale Image	8	256	Shades of grey from 0 (black) to 255 (white).
True Colour (RGB)	24	16.7M	Three 8-bit channels: Red, Green, Blue.
Pseudo Colour	8	256	Greyscale mapped to colours using a lookup table.
Monochromatic	1-8	2-256	Single hue; intensity varies. Subset of greyscale.

UNIT 2: Image Transforms (DFT, DCT, Haar, Wavelet)

DFT — All 7 Semesters ★★★ | DCT — 3 Semesters ★★ | Haar — 3 Semesters ★★ | Wavelet — 5 Semesters ★★★

2.1 Discrete Fourier Transform (DFT)

DFT converts an image from the spatial domain to the frequency domain. It represents the image as a sum of sinusoidal components of different frequencies.

```
1-D DFT:
F(u) = Σ[x=0 to N-1] f(x) · e-j2πux/N      u = 0, 1, ..., N-1

Inverse 1-D DFT:
f(x) = (1/N) Σ[u=0 to N-1] F(u) · ej2πux/N

2-D DFT (for images of size M×N):
F(u,v) = Σ Σ f(x,y) · e-j2π(ux/M + vy/N)

2-D Inverse DFT:
f(x,y) = (1/MN) Σ Σ F(u,v) · ej2π(ux/M + vy/N)
```

Properties of DFT

Property	Statement
Linearity	DFT[a·f + b·g] = a·F + b·G
Periodicity	F(u,v) = F(u+M, v+N)
Conjugate Symmetry	F(u,v) = F*(-u,-v) for real images
Translation	Shift in space → phase change in frequency; shift in frequency → phase in space
Rotation	Rotating f(x,y) by θ rotates F(u,v) by the same θ
Separability	2-D DFT can be computed as two successive 1-D DFTs
Convolution Theorem	Convolution in space = Multiplication in frequency
Parseval's Theorem	Energy is preserved: Σ f ² = (1/MN)Σ F ²

Solved — 1-D DFT Computation (Jun 2024)

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Problem: Compute DFT of f(x) = [1, 0, -1, 0] (N=4)
F(u) = Σ f(x) e-j2πux/4 for u = 0,1,2,3
```

$$F(0) = f(0)e^{j0} + f(1)e^{j0} + f(2)e^{j0} + f(3)e^{j0} \\ = 1 + 0 + (-1) + 0 = 0$$

$$F(1) = f(0)e^{j0} + f(1)e^{-j\pi/2} + f(2)e^{-j\pi} + f(3)e^{-j3\pi/2} \\ = 1 + 0 \cdot (-j) + (-1)(-1) + 0 \cdot (j) = 1 + 0 + 1 + 0 = 2$$

$$F(2) = 1 + 0 \cdot (-1) + (-1) \cdot 1 + 0 \cdot (-1) = 1 - 1 = 0$$

$$F(3) = 1 + 0 \cdot (j) + (-1) \cdot (-1) + 0 \cdot (-j) = 1 + 1 = 2$$

Result: $F = [0, 2, 0, 2]$

Solved — 2-D DFT (Dec 2023)

Problem: Compute 2-D DFT of $f(x,y) = [[1,1],[1,1]]$ (2×2 image)

$$F(u,v) = \sum_x \sum_y f(x,y) \cdot e^{-j2\pi(ux/2 + vy/2)}$$

$$F(0,0) = f(0,0) + f(0,1) + f(1,0) + f(1,1) \\ = 1 + 1 + 1 + 1 = 4$$

$$F(0,1) = f(0,0) \cdot e^{j0} + f(0,1) \cdot e^{-j\pi} + f(1,0) \cdot e^{j0} + f(1,1) \cdot e^{-j\pi} \\ = 1 \cdot 1 + 1 \cdot (-1) + 1 \cdot 1 + 1 \cdot (-1) = 0$$

$$F(1,0) = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot (-1) + 1 \cdot (-1) = 0$$

$$F(1,1) = 1 \cdot 1 + 1 \cdot (-1) + 1 \cdot (-1) + 1 \cdot 1 = 0$$

Result: $F(u,v) = [[4, 0], [0, 0]]$

2.2 Discrete Cosine Transform (DCT)

DCT represents an image as a sum of cosine functions. It is widely used in JPEG compression as it concentrates most signal energy in few coefficients.

1-D DCT of order N:

$$C(u) = \alpha(u) \cdot \sum_{x=0}^{N-1} f(x) \cdot \cos[(2x+1)u\pi / 2N]$$

$$\text{where } \alpha(0) = \sqrt{1/N}, \quad \alpha(u) = \sqrt{2/N} \text{ for } u \neq 0$$

DCT Matrix of Order 2 (computed from formula):

$$T = (1/\sqrt{2}) \times \begin{bmatrix} 1 & 1 \\ \cos(\pi/4) & -\cos(\pi/4) \end{bmatrix} \\ = \begin{bmatrix} 0.707 & 0.707 \\ 0.707 & -0.707 \end{bmatrix}$$

Advantages of DCT over DFT

- DCT uses only real numbers; DFT involves complex numbers — DCT is computationally cheaper.
- DCT achieves better energy compaction — most image energy in fewer coefficients.
- DCT avoids boundary discontinuity (Gibbs phenomenon) better than DFT.
- DCT is the basis of JPEG, MPEG compression standards.

2.3 Haar Transform

The Haar transform is the simplest wavelet transform. It uses Haar basis functions (rectangular pulses) and is computationally very fast.

```
Haar matrix for N=2:
H = (1/sqrt(2)) * [[1, 1],
                  [1, -1]]

Transform: T = H · f · H^T
Inverse:   f = H^T · T · H
```

Solved — Haar Transform of 2×2 matrix (Dec 2022)

Given $f = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$$H = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Step 1: $H \cdot f$

$$\text{Row1} = (1/\sqrt{2})[1 \times 1 + 1 \times 3, 1 \times 2 + 1 \times 4] = (1/\sqrt{2})[4, 6]$$

$$\text{Row2} = (1/\sqrt{2})[1 \times 1 - 1 \times 3, 1 \times 2 - 1 \times 4] = (1/\sqrt{2})[-2, -2]$$

Step 2: $(H \cdot f) \cdot H^T$

$$= (1/2)[[4 \times 1 + 6 \times 1, 4 \times 1 + 6 \times (-1)],$$

$$[-2 \times 1 + (-2) \times 1, -2 \times 1 + (-2) \times (-1)]]$$

$$= (1/2)[[10, -2], [-4, 0]]$$

$$= [[5, -1], [-2, 0]]$$

2.4 Wavelet Transform

Wavelets are small oscillatory waveforms that are localised in both time and frequency. Unlike DFT which gives only frequency info, wavelets give frequency AND location information.

Feature	Fourier Transform	Wavelet Transform
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Basis functions	Sinusoids (infinite extent)	Wavelets (finite extent)
Frequency localisation	Yes	Yes
Time/space localisation	No	Yes
Multi-resolution	No	Yes
Best for	Stationary signals	Non-stationary / transient signals
Compression	Good (DCT for JPEG)	Better (used in JPEG2000)

Feature	CWT (Continuous)	DWT (Discrete)
Scale	Continuous values	Discrete (dyadic) values
Redundancy	Highly redundant	No redundancy
Computation	Expensive	Fast ($O(N)$)
Application	Analysis, theory	Compression, denoising
Reconstruction	Possible	Exact reconstruction

Solved — Orthogonal Transform (Dec 2023)

Given: $A = (1/\sqrt{2})[[1, 1], [1, -1]]$, $f = [[1, 3], [5, 7]]$

Forward: $F = A \cdot f \cdot A^T$

$A \cdot f$:

$$\text{Row1} = (1/\sqrt{2})[1+5, 3+7] = (1/\sqrt{2})[6, 10]$$

$$\text{Row2} = (1/\sqrt{2})[1-5, 3-7] = (1/\sqrt{2})[-4, -4]$$

$(A \cdot f) \cdot A^T$:

$$= (1/2)[[6+10, 6-10], [-4-4, -4+4]]$$

$$= (1/2)[[16, -4], [-8, 0]]$$

$$= [[8, -2], [-4, 0]]$$

Inverse: $f_{\text{rec}} = A^T \cdot F \cdot A = [[1, 3], [5, 7]]$ (exact recovery)

UNIT 3: Image Enhancement

Appeared in: ALL 7 Semesters | Priority: ★★★

3.1 Introduction

Image enhancement improves the visual quality of an image for better perception or subsequent processing. It is a subjective process — the 'best' result depends on the application.

Spatial Domain	Frequency Domain
Directly operates on pixels $f(x,y)$	Operates on Fourier transform $F(u,v)$
Simple and fast	More powerful for global operations
Point processing: histogram eq., contrast stretch	Filtering: LPF, HPF, Bandpass
Mask/spatial filtering: smoothing, sharpening	Wiener filter, Homomorphic filter

3.2 Spatial Domain Techniques

A) Histogram Equalization (All 7 Semesters ★★★)

Histogram equalization redistributes pixel intensities to produce a flat histogram, improving contrast in images with narrow intensity ranges.

Steps:

1. Compute histogram $h(r_k)$ = number of pixels with intensity r_k
2. Compute normalized histogram: $p(r_k) = h(r_k) / (M \times N)$
3. Compute CDF: $s_k = (L-1) \times \sum_{j=0}^k p(r_j)$
4. Map each pixel: new intensity = round(s_k)

Why it gives flat histogram:

The transformation $s = T(r) = (L-1) \int_0^r p(w)dw$
is chosen so that the output PDF $p_s(s) = 1/(L-1)$ — uniform/flat.

B) Histogram Matching (Specification) (Jun 2024 ★★)

Histogram matching transforms an image so its histogram matches a specified (desired) histogram. Useful when the target distribution is known.

1. Equalise the source image: $s = T(r)$
2. Equalise the desired histogram: $v = G(z)$
3. Apply inverse mapping: $z = G^{-1}(s)$

C) Contrast Stretching (Dec 2022 ★★)

Contrast stretching expands the range of intensity values in an image to span the full dynamic range [0, L-1].

$$s = (r - r_{\min}) \times [(L-1) / (r_{\max} - r_{\min})]$$

Where: r = input intensity
 $r_{\min}, r_{\max}$ = actual min/max in image
 L = number of grey levels (256 for 8-bit)

D) Spatial Filtering — Smoothing (Low-Pass)

Filter	Description
Mean Filter (Average)	Replaces each pixel with the average of its neighbours. Reduces noise but blurs edges. Kernel: $(1/9) \times [[1,1,1],[1,1,1],[1,1,1]]$
Weighted Average Filter	Centre pixel given more weight. Less blurring than simple average.
Median Filter	Replaces pixel with median of neighbourhood. Excellent for salt-and-pepper noise. Preserves edges better than mean filter.
Gaussian Filter	Uses Gaussian function as kernel weights. Smooth noise reduction. No ringing. Kernel values: $G(x,y) = e^{-(x^2+y^2)/2\sigma^2}$

E) Laplacian Filter — Image Sharpening (Dec 2024 / Jun 2024 ★★★)

The Laplacian is a second-derivative operator that highlights regions of rapid intensity change (edges). It is used for image sharpening.

Laplacian of $f(x,y)$:

$$\nabla^2 f = \partial^2 f / \partial x^2 + \partial^2 f / \partial y^2$$

Discrete approximation:

$$\nabla^2 f(x,y) = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y)$$

Laplacian Mask (4-neighbour):

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Laplacian Mask (8-neighbour):

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Sharpened image:

$$g(x,y) = f(x,y) - \nabla^2 f(x,y) \quad (\text{when centre is negative})$$

$$g(x,y) = f(x,y) + \nabla^2 f(x,y) \quad (\text{when centre is positive})$$

3.3 Frequency Domain Enhancement

Filtering in the frequency domain is done by multiplying the Fourier transform of the image by a filter transfer function $H(u,v)$.

Steps for frequency domain filtering:

1. Compute $F(u,v) = \text{DFT}[f(x,y)]$
2. Shift zero-frequency to centre: multiply $f(x,y)$ by $(-1)^{(x+y)}$
3. Multiply: $G(u,v) = H(u,v) \times F(u,v)$
4. Shift back
5. Compute inverse DFT: $g(x,y) = \text{IDFT}[G(u,v)]$
6. Take real part

Filter	Ideal LPF	Butterworth LPF	Gaussian LPF
Transfer Function $H(u,v)$	1 if $D \leq D_0$, 0 if $D > D_0$	$1/[1+(D/D_0)^{2n}]$	$e^{-(D^2/2D_0^2)}$
Ringing Effect	Severe (Gibbs)	Moderate (high order)	None
Transition	Sharp cutoff	Gradual	Very smooth
Order parameter	—	n (controls rolloff)	—
Application	Theoretical only	Practical smoothing	Image denoising

UNIT 4: Noise Models & Image Restoration

Appeared in: ALL 7 Semesters | Priority: ★★★

4.1 Noise Models

Noise is random variation of brightness or colour in images. It originates from image sensors, transmission errors, or environmental interference.

Noise Type	PDF / Description
Gaussian Noise	Symmetric bell curve. PDF: $p(z) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-(z-\mu)^2/2\sigma^2}$. Most common; results from thermal fluctuations. Mean μ , variance σ^2 .
Salt & Pepper (Impulse)	Random bright (255) and dark (0) pixels. Caused by sudden disturbances or bit errors. Best removed by median filter.
Rayleigh Noise	Skewed distribution. PDF: $p(z) = (2z/b) \cdot e^{-z^2/b}$ for $z \geq 0$. Common in radar imaging.
Gamma (Erlang) Noise	PDF: $p(z) = (a^b \cdot z^{b-1} \cdot e^{-az}) / (b-1)!$ for $z \geq 0$. Used in laser imaging.
Exponential Noise	Special case of Gamma ($b=1$). PDF: $p(z) = a \cdot e^{-az}$ for $z \geq 0$.
Uniform Noise	Constant PDF between a and b . PDF: $p(z) = 1/(b-a)$. Mean = $(a+b)/2$.
Periodic Noise	Regular pattern of interference. Appears as spikes in frequency domain. Best removed by bandreject filter.

4.2 Image Degradation Model

Degradation model:

$$g(x, y) = h(x, y) * f(x, y) + \eta(x, y)$$

In frequency domain:

$$G(u, v) = H(u, v) \cdot F(u, v) + N(u, v)$$

Where: $f(x, y)$ = original image

$h(x, y)$ = degradation function (PSF)

$\eta(x, y)$ = additive noise

$g(x, y)$ = degraded (observed) image

4.3 Inverse Filtering

Inverse filtering attempts to recover the original image by dividing by the degradation function in the frequency domain.

$$\begin{aligned}\hat{F}(u,v) &= G(u,v) / H(u,v) \\ &= F(u,v) + N(u,v)/H(u,v)\end{aligned}$$

Limitations:

1. If $H(u,v) = 0$ at any point $\rightarrow$ division by zero $\rightarrow$ undefined
2. Noise term $N(u,v)/H(u,v)$ is amplified enormously when H is small
3. Not practical without noise – always adds noise amplification

4.4 Wiener Filter (Dec 2025 ★★)

The Wiener filter is an optimal linear filter that minimises the mean square error between the estimated and the true image. It balances restoration and noise suppression.

Wiener Filter Transfer Function:

$$\hat{H}_w(u,v) = H^*(u,v) / [|H(u,v)|^2 + S_\eta(u,v)/S_f(u,v)]$$

or equivalently:

$$\hat{H}_w(u,v) = [1/H(u,v)] \times [|H(u,v)|^2 / (|H(u,v)|^2 + K)]$$

Where: $H(u,v)$ = degradation function

$H^*(u,v)$ = complex conjugate of H

$S_\eta(u,v)$ = power spectral density of noise

$S_f(u,v)$ = power spectral density of original image

$K = S_\eta/S_f$ = noise-to-signal power ratio (constant approximation)

Problems it solves:

1. Avoids excessive noise amplification (unlike inverse filter)
2. Minimises MSE between restored and original image
3. Handles both noise and blur simultaneously

UNIT 5: Colour Models

Appeared in: 6/7 Semesters | Priority: ★★★

5.1 RGB Colour Model

The RGB model is additive — colours are created by adding Red, Green, Blue primary lights. Used in monitors, cameras, and displays.

Colour cube: each axis represents R, G, or B (0 to 255 for 8-bit)

Black = (0, 0, 0) White = (255, 255, 255)

Red = (255, 0, 0) Green = (0, 255, 0)

Blue = (0, 0, 255) Yellow = (255, 255, 0)

Cyan = (0, 255, 255) Magenta = (255, 0, 255)

Grey levels in primary images:

On red face (R=255): Greens vary 0-255, Blues vary 0-255

On green face (G=255): Reds vary 0-255, Blues vary 0-255

5.2 CMY and CMYK Models

CMY is subtractive — used in colour printing. CMYK adds a black (K) channel to improve print quality and reduce ink cost.

RGB to CMY conversion:

$C = 1 - R$

$M = 1 - G$

$Y = 1 - B$

CMY to CMYK conversion (K = under colour removal):

$K = \min(C, M, Y)$

$C' = (C - K) / (1 - K)$

$M' = (M - K) / (1 - K)$

$Y' = (Y - K) / (1 - K)$

5.3 HSI / HSV Model

HSI (Hue, Saturation, Intensity) separates colour information (H, S) from intensity (I). Very useful for image processing as it matches human colour perception.

Component	Description
Hue (H)	The dominant colour wavelength; described by an angle 0°-360° on a colour wheel. Red=0°, Green=120°, Blue=240°.

Saturation (S)	The purity of the colour. S=0 means grey. S=1 means fully saturated (pure) colour.
Intensity (I)	The overall brightness. I=0 is black, I=1 is white. Independent of colour.

RGB to HSI:

$$I = (R + G + B) / 3$$

$$S = 1 - (3 / (R+G+B)) \times \min(R, G, B)$$

$$H = \arccos\{ [(R-G)/2 + (R-B)/2] / \sqrt{[(R-G)^2 + (R-B)(G-B)]} \}$$

(if $B > G$, then $H = 360^\circ - H$)

5.4 Colour Terms

Term	Definition
Brightness	Achromatic notion of intensity — how light or dark an image appears. Ranges from black to white.
Contrast	Difference in luminance between the brightest and darkest regions. High contrast = clear distinction between light and dark areas.
Hue	The attribute associated with the dominant wavelength. Describes the 'type' of colour (red, green, blue, etc.).
Saturation	The degree to which a colour is diluted by white light. A fully saturated colour contains no white.
Chromaticity	A combination of Hue and Saturation. Describes the quality/purity of colour independent of intensity.
Intensity	The overall amplitude of the light. Combined R+G+B components.

Solved — Colour Component Calculation (Dec 2024)

Given: warm white (x=0.45, y=0.4), deep blue (x=0.15, y=0.2)

Chromaticity coordinates: $x + y + z = 1 \Rightarrow z = 1 - x - y$

Warm white: $z = 1 - 0.45 - 0.4 = 0.15$

$(x, y, z) = (0.45, 0.40, 0.15) \Rightarrow R=45\%, G=40\%, B=15\%$

Deep blue: $z = 1 - 0.15 - 0.2 = 0.65$

$(x, y, z) = (0.15, 0.20, 0.65) \Rightarrow R=15\%, G=20\%, B=65\%$

UNIT 6: Clustering Methods

K-Means: 6/7 Semesters ★★★ | Hierarchical: 5/7 Semesters ★★★

6.1 Introduction to Clustering

Clustering is an unsupervised learning technique that groups similar data points together without any prior labelling. The goal is to maximize intra-cluster similarity and minimise inter-cluster similarity.

Category	Methods
Partition-based	K-Means, K-Medoids, Fuzzy C-Means
Hierarchical	Agglomerative (bottom-up), Divisive (top-down)
Density-based	DBSCAN, OPTICS
Grid-based	STING, CLIQUE
Model-based	EM Algorithm, Gaussian Mixture Models

6.2 K-Means Clustering (All 6 Papers ★★★)

K-Means partitions N data points into K clusters by minimising the total within-cluster sum of squared distances from each point to its cluster centroid.

Algorithm

- Choose K (number of clusters)
- Randomly initialise K cluster centroids $\mu_1, \mu_2, \dots, \mu_k$
- Assignment step: Assign each point x_i to cluster $j = \operatorname{argmin} ||x_i - \mu_j||^2$
- Update step: Recalculate centroids: $\mu_j = (1/|C_j|) \sum_{i \in C_j} x_i$
- Repeat steps 3-4 until centroids no longer change (convergence)

Distance measures used in K-Means:

Euclidean: $d(x, y) = \sqrt{\sum (x_i - y_i)^2}$
 Manhattan: $d(x, y) = \sum |x_i - y_i|$
 Minkowski: $d(x, y) = [\sum |x_i - y_i|^p]^{1/p}$
 Chebyshev: $d(x, y) = \max |x_i - y_i|$

Objective function to minimise:

$$J = \sum_j \sum_{i \in C_j} ||x_i - \mu_j||^2$$

Example — K-Means (Jun 2024 / Jun 2025)

Given points: A(1,1), B(2,1), C(4,3), D(5,4), E(1,2) — K=2

Initial centroids: $\mu_1=A(1,1)$, $\mu_2=D(5,4)$

Iteration 1 — Assignment:

B(2,1): $d(\mu_1)=1.0$, $d(\mu_2)=5.0 \rightarrow$ Cluster 1

C(4,3): $d(\mu_1)=3.61$, $d(\mu_2)=1.41 \rightarrow$ Cluster 2

E(1,2): $d(\mu_1)=1.0$, $d(\mu_2)=5.1 \rightarrow$ Cluster 1

Cluster 1: {A,B,E} Cluster 2: {C,D}

Iteration 1 — Update:

$\mu_1 = ((1+2+1)/3, (1+1+2)/3) = (1.33, 1.33)$

$\mu_2 = ((4+5)/2, (3+4)/2) = (4.5, 3.5)$

Continue until no change in cluster assignments.

6.3 Hierarchical Clustering (Dec 2025, Dec 2023 ★★★)

Hierarchical clustering builds a tree (dendrogram) of nested clusters. No need to specify K beforehand.

Feature	Agglomerative (Bottom-up)	Divisive (Top-down)
Start	Each point is its own cluster	All points in one cluster
Process	Merge closest pair iteratively	Split largest cluster iteratively
Result	Dendrogram built from leaves up	Dendrogram built from root down
Complexity	$O(n^2 \log n)$	$O(2^n)$ exact, $O(n^2)$ approx.
Common use	Most widely used	Less common

Agglomerative Algorithm Steps

9. Compute distance matrix D for all pairs of points
10. Find the pair (i,j) with minimum distance
11. Merge clusters i and j into one new cluster
12. Update distance matrix using linkage method
13. Repeat until only one cluster remains

Linkage Type	Distance Formula	Characteristic
Single Link (Min)	$d(A,B) = \min \text{ dist between any point in A and any in B}$	Chaining effect; elongated clusters
Complete Link (Max)	$d(A,B) = \max \text{ dist between any point in A and any in B}$	Compact, roughly equal clusters

Average Link	$d(A,B)$ = avg of all pairwise distances between A and B	Balance between single and complete
Centroid Link	$d(A,B)$ = distance between centroids of A and B	Computationally simple

6.4 Distance Measures for Clustering

Distance Measure	Formula / Description
Euclidean	$d = \sqrt{\sum (x_i - y_i)^2}$. The straight-line distance. Most common.
Manhattan (City Block)	$d = \sum x_i - y_i $. Sum of absolute differences. Less sensitive to outliers.
Mahalanobis	$d = \sqrt{(x-y)^T \Sigma^{-1} (x-y)}$. Accounts for covariance. Scale-invariant.
Cosine Similarity	$\cos(\theta) = (x \cdot y) / (x y)$. Measures angle between vectors. Used in text.
Hamming	Number of positions where two binary strings differ. Used for categorical data.

UNIT 7: Supervised Learning & Classification

Appeared in: 6/7 Semesters | Priority: ★★★

7.1 Supervised vs Unsupervised Learning

Aspect	Supervised Learning	Unsupervised Learning
Definition	Learns from labelled training data	Finds patterns in unlabelled data
Output	Predicts label/value for new inputs	Discovers groupings or structure
Data requirement	Needs labelled pairs (x, y)	Only feature vectors x needed
Feedback	Error signal available	No direct error signal
Examples	SVM, Decision Trees, Neural Nets, Naive Bayes	K-Means, Hierarchical, DBSCAN
Applications	Object recognition, spam detection	Customer segmentation, anomaly detection
Evaluation	Accuracy, precision, recall, F1	Silhouette score, Davies-Bouldin index

7.2 Categories of Supervised Learning Algorithms

Category	Description & Examples
Instance-based (Lazy)	No explicit training phase. Memorises all training examples. Classifies new input by comparing to stored examples. Example: K-Nearest Neighbour (KNN).
Decision Tree	Builds a tree structure of feature tests. Each internal node = test on a feature. Each leaf = class label. Example: ID3, C4.5, CART.
Support Vector Machine (SVM)	Finds the optimal hyperplane that maximises the margin between classes. Uses kernel trick for non-linear boundaries. Effective in high dimensions.
Neural Networks / Deep Learning	Multi-layer networks of neurons. Learn hierarchical feature representations. CNN for images, RNN for sequences. Revolutionised object recognition (ImageNet 2012).
Naive Bayes / Bayesian	Probabilistic classifier based on Bayes theorem. Assumes feature independence. Fast and effective for text and image classification.
Linear Discriminant Analysis (LDA)	Finds linear combination of features that best separates classes. Maximises between-class variance while minimising within-class variance.

7.3 Linear Discriminant Function

A Linear Discriminant Function (LDF) is a linear function of the feature vector used to classify patterns into different classes.

For 2-class problem:

$$g(x) = w^T x + w_0$$

If $g(x) > 0 \rightarrow$ assign to class ω_1

If $g(x) < 0 \rightarrow$ assign to class ω_2

If $g(x) = 0 \rightarrow$ on the decision boundary

Where: w = weight vector (normal to decision hyperplane)

w_0 = bias/threshold weight

x = feature vector

For multi-class (one-vs-rest):

$$g_i(x) = w_i^T x + w_{0i} \quad \text{for } i = 1, \dots, C$$

Assign to class $i^* = \operatorname{argmax}_i g_i(x)$

Properties of LDA

- The decision boundary is a hyperplane (line in 2D, plane in 3D).
- Maximises ratio of between-class scatter to within-class scatter (Fisher Criterion).
- Assumes Gaussian class-conditional distributions with equal covariance.
- Reduces dimensionality: N classes $\rightarrow$ at most $N-1$ discriminant functions.
- Equivalent to minimum Euclidean distance classifier when covariances are equal.

7.4 Bayesian Classifier

Bayesian classification uses Bayes theorem to compute the posterior probability of each class given the observed features.

Bayes Theorem:

$$P(\omega_j | x) = P(x | \omega_j) \times P(\omega_j) / P(x)$$

$$\text{Posterior} = \text{Likelihood} \times \text{Prior} / \text{Evidence}$$

Bayes Decision Rule (minimum error):

Assign x to class ω_i if $P(\omega_i | x) > P(\omega_j | x)$ for all $j \neq i$

Naive Bayes (independence assumption):

$$P(x | \omega_j) = \prod P(x_i | \omega_j) \quad (\text{product of individual feature likelihoods})$$

Properties of Bayesian Classifier

- Optimal: minimises probability of classification error.
- Requires knowledge of class-conditional PDFs (often assumed Gaussian).

- Equivalent to minimum Euclidean distance classifier under equal covariance, equal priors.
- Naive Bayes is computationally cheap and surprisingly effective in practice.

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UNIT 8: Computer Vision — Camera Models, Edge Detection & Segmentation

Camera Models: 4 Semesters ★★ | Edge Detection: 4 Semesters ★★ | Segmentation: 4 Semesters ★★

8.1 Camera Models

Pin-Hole Camera Model (Jun 2025 ★★)

The pin-hole camera model is the simplest model of image formation. Light from a 3D scene passes through a tiny hole (aperture) and forms an inverted image on the image plane.

Perspective projection equations:

$$u = f \times X/Z$$

$$v = f \times Y/Z$$

Where: (X, Y, Z) = 3D world coordinates

(u, v) = 2D image coordinates

f = focal length

Matrix form (homogeneous):

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Camera Calibration (Dec 2025 ★★)

Camera calibration determines the intrinsic parameters (focal length, principal point, distortion) and extrinsic parameters (rotation and translation) of a camera.

Parameter Type	Parameters
Intrinsic	Focal length (fx, fy), Principal point (cx, cy), Skew coefficient, Radial/tangential distortion coefficients
Extrinsic	Rotation matrix R (3×3), Translation vector t (3×1) — describes camera position and orientation in world coordinates

Single vs Multiple Camera Systems

Feature	Single Camera	Multiple Cameras (Stereo)
Depth perception	Limited (monocular depth cues)	Direct (triangulation)
3D reconstruction	Requires motion or assumptions	Direct stereo matching
Cost	Low	Higher (multiple sensors)

Calibration	Intrinsic only	Intrinsic + extrinsic (baseline)
Applications	Object detection, tracking	Autonomous vehicles, robotics, VR

8.2 Edge Detection

Edge detection identifies boundaries of objects in images. Edges occur where there is a rapid change in image intensity.

Detector	Method & Properties
Sobel	Uses 3×3 masks for G_x and G_y . G_x detects vertical edges, G_y detects horizontal. Magnitude = $\sqrt{G_x^2 + G_y^2}$. Simple and noise-resistant.
Prewitt	Similar to Sobel but with equal weights. Less noise resistance than Sobel.
Roberts Cross	2×2 diagonal gradient computation. Fast but very sensitive to noise.
Laplacian of Gaussian (LoG)	Smooths with Gaussian first then applies Laplacian. Detects zero crossings. Good for fine edges. Sensitive to noise if Gaussian σ is small.
Canny Edge Detector	Multi-stage: (1) Gaussian smoothing, (2) Gradient computation, (3) Non-maximum suppression, (4) Double thresholding, (5) Edge tracking by hysteresis. Optimal: best detection, localisation, and minimal false edges.

8.3 Image Segmentation

Segmentation partitions an image into meaningful regions. Each region should be homogeneous with respect to some property (intensity, colour, texture).

Method	Description
Thresholding	Simplest method. Pixels above threshold → foreground; below → background. Global or adaptive. Fails with uneven lighting.
Edge-based Segmentation	Uses edge detectors to find boundaries, then links edges into closed regions. Suitable when boundaries are well defined.
Region-based Segmentation	Region growing: start from seed pixels and expand. Region splitting and merging. Good for uniform regions.
Clustering-based	K-means or fuzzy clustering applied to pixel feature vectors (intensity, colour, texture). Scalable to large images.
Graph-based	Image as graph; pixels = nodes, edges = similarity. Minimum cut partitions graph. Normalised cuts (Shi & Malik).

8.4 Homography (Dec 2025 ★)

Homography is a projective transformation that maps points from one image plane to another. It is a 3×3 matrix H (up to scale, 8 degrees of freedom).

Homography relation:

$$x' = H \cdot x \quad (\text{in homogeneous coordinates})$$

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

Applications:

- Image stitching / panorama creation
- Perspective correction / rectification
- Augmented reality (AR overlay)
- Camera pose estimation
- Document scanning (de-warping)

8.5 Feature Extraction

Feature extraction transforms raw image data into a compact set of discriminative features for recognition and classification.

Feature Type	Description & Examples
Colour Features	Colour histograms, colour moments, dominant colour. Compact but not spatial.
Texture Features	GLCM (Grey Level Co-occurrence Matrix), Gabor filters, LBP. Captures surface patterns.
Shape Features	Hu moments, Fourier descriptors, contour features. Describes object outline.
Local Features (Keypoints)	SIFT, SURF, ORB. Scale and rotation invariant. Used for matching.
Deep Features	CNN activations from trained networks. Best performance on recognition tasks.

UNIT 9: Short Notes on High-Frequency Topics

These topics appear repeatedly in the 'short notes' question (Q5) every semester

9.1 Gaussian Noise

Gaussian noise has a probability density function following the Gaussian (normal) distribution. It is the most common type of noise in digital images, arising from thermal fluctuations in image sensors.

PDF: $p(z) = (1 / \sqrt{2\pi\sigma^2}) \times e^{-(z-\mu)^2 / 2\sigma^2}$

Parameters: μ = mean (often 0), σ^2 = variance

Removal: Gaussian smoothing filter, Wiener filter

Characteristic: Symmetric bell shape; independent at each pixel

9.2 Salt and Pepper Noise

Salt and pepper (impulse) noise randomly sets some pixels to the maximum value (white/salt) or minimum value (black/pepper). Caused by sudden intense disturbances.

PDF: $p(z) = \begin{cases} P_a & \text{if } z = a \text{ (pepper, dark)} \\ P_b & \text{if } z = b \text{ (salt, bright)} \\ 1-P_a-P_b & \text{otherwise} \end{cases}$

Best filter: Median filter (preserves edges, removes impulse noise)

9.3 Median Filter

The median filter replaces each pixel with the median value of the intensities in its neighbourhood. It is a non-linear filter, excellent for removing impulse noise while preserving edges.

Algorithm: Sort neighbourhood pixel values → pick middle value (median) → replace centre pixel.

9.4 Gaussian Low-Pass Filter (GLPF)

The Gaussian LPF removes high-frequency components (noise, edges) while keeping low-frequency components (smooth regions). The filter has no sharp cutoff so there is no ringing.

Spatial domain: $G(x,y) = (1/2\pi\sigma^2) \times e^{-(x^2+y^2)/2\sigma^2}$

Frequency domain: $H(u,v) = e^{-D(u,v)^2/2D_0^2}$

Where $D(u,v)$ = distance from centre, D_0 = cutoff frequency

9.5 Butterworth Low-Pass Filter

$$H(u,v) = 1 / [1 + (D(u,v)/D_0)^{2n}]$$

n = filter order (higher $n \rightarrow$ sharper cutoff)

No ringing for low n ; some ringing at high n

9.6 Ideal Low-Pass Filter (ILPF)

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \leq D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

Sharp cutoff causes severe ringing (Gibbs phenomenon) – not practical.

9.7 Band-Pass and Band-Reject Filters

Band-pass filters pass a specific frequency band. Band-reject (notch) filters block a specific band. Used to remove periodic noise (appears as spikes in frequency domain).

9.8 Pixel Resolution vs Spatial Resolution vs Intensity Resolution

Term	Definition
Pixel Resolution	Total number of pixels in an image: Width $\times$ Height (e.g., $1920 \times 1080 = 2,073,600$ pixels = 2 MP)
Spatial Resolution	Number of pixels per unit area or length (PPI or DPI). Higher PPI = finer detail visible.
Intensity Resolution	Number of distinct intensity levels (grey levels). Determined by bit depth. 8-bit = 256 levels.

9.9 Stereovision (Jun 2025 ★★)

Stereovision uses two cameras (like human eyes) to compute depth by triangulation. The horizontal shift of a point between left and right images is called disparity.

Disparity: $d = x_{\text{left}} - x_{\text{right}}$

Depth: $Z = (f \times B) / d$

Where: f = focal length, B = baseline (camera separation)

d = disparity (larger disparity $\rightarrow$ closer object)

9.10 Pseudo Colour Images

Pseudo colour assigns colours to greyscale image intensities using a lookup table (LUT). The colours carry no physical meaning but help human visual interpretation. Common in medical imaging, thermal imaging.

9.11 K-Means Clustering (short notes version)

K-Means partitions data into K groups by iteratively assigning points to nearest centroid and updating centroids. Sensitive to initial centroid choice and outliers. Time complexity $O(n \cdot K \cdot I \cdot d)$ where I = iterations, d = dimensions.

9.12 Supervised Learning (short notes version)

Supervised learning uses labelled training data (x, y) to learn a mapping $f(x) \rightarrow y$. The model is trained to minimize prediction error. After training, it generalises to unseen data. Applications: image classification, spam detection, medical diagnosis.

9.13 Haar Transform (short notes)

The Haar transform uses box-like basis functions, is the simplest wavelet transform, computed via average and difference operations, has $O(N)$ complexity, and is used in image compression (JPEG2000) and face detection (Viola-Jones algorithm).

9.14 Cosine Transformation (DCT) — Short Note

DCT expresses a signal as a sum of cosine waves of different frequencies. Used in JPEG/MPEG compression. Achieves high energy compaction. Works on real numbers only. Closely related to DFT but avoids complex exponentials.

9.15 Region Detection

Region detection identifies connected areas in an image with homogeneous properties. Used in scene understanding, object recognition, and medical image analysis. Methods: region growing, watershed algorithm, graph cuts.

9.16 Edge-Based Segmentation

Uses edge detectors (Sobel, Canny) to find sharp intensity changes, then links detected edges into closed boundary contours to segment regions. Challenges: gaps in edges, noise sensitivity. Applications: autonomous driving lane detection, object counting.

UNIT 10: All Numerical Problems — Quick Reference

10.1 Image Size Calculations

Formula sheet

1. Number of samples = width_inches $\times$ DPI $\times$ height_inches $\times$ DPI
2. Given aspect ratio a:b and total pixels P:
 $k_a \times k_b = P \Rightarrow k = \sqrt{P / (a \times b)}$
 Width = $a \times k$, Height = $b \times k$
3. Image storage (bytes) = Width $\times$ Height $\times$ bits_per_pixel / 8
4. PPI = diagonal_pixels / diagonal_inches
 $\text{diagonal_pixels} = \sqrt{W^2 + H^2}$
5. Pixel density (medical): cycles/mm $\times$ mm/inch = cycles/inch
 pixels needed per cycle = sampling_rate / cycles_per_inch

10.2 DFT Problems — All Past Numerical

$f(x) = [1, 0, -1, 0]$ N=4 (Jun 2024)

$F(0) = 0, F(1) = 2, F(2) = 0, F(3) = 2$

2-D DFT of all-ones 2×2 matrix (Dec 2023)

$f = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow F = \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix}$

2-D DFT of identity-like 2×2 (construct DFT matrix N=2 — Dec 2022)

$W_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ (DFT matrix of order 2)

$F = W_2 \cdot f \cdot W_2^*$

10.3 Orthogonal Transform — Dec 2023

$A = (1/\sqrt{2})\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, f = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}$

$F = A \cdot f \cdot A^T = \begin{bmatrix} 8 & -2 \\ -4 & 0 \end{bmatrix}$

Inverse: $f = A^T \cdot F \cdot A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix} \checkmark$

10.4 DCT Matrix Order 2

Standard 2×2 DCT matrix

$$T[0,0] = \sqrt{1/2} = 0.7071$$

$$T[0,1] = \sqrt{1/2} = 0.7071$$

$$T[1,0] = \cos(\pi/4) = 0.7071$$

$$T[1,1] = \cos(3\pi/4) = -0.7071$$

$$T = \begin{bmatrix} 0.7071 & 0.7071 \\ 0.7071 & -0.7071 \end{bmatrix}$$

10.5 Haar Transform N=2 — Dec 2022

$$H = (1/\sqrt{2})[[1,1],[1,-1]], f = [[1,2],[3,4]]$$

$$T = H \cdot f \cdot H^T = [[5,-1],[-2,0]]$$

10.6 Medical Image Sampling — Dec 2022**8×8 inch medical image, sampling = 5 cycles/mm — how many pixels per cycle?**

Sampling resolution = 5 cycles/mm × 25.4 mm/inch = 127 cycles/inch

For Nyquist: need ≥ 2 pixels per cycle

Pixels per inch needed = $2 \times 127 = 254$ pixels/inch

For 8-inch image: $8 \times 254 = 2032$ pixels per side

256×256 image $\rightarrow 256/8 = 32$ pixels/inch $\rightarrow 32/(2 \times 127) < 1 \rightarrow$ NOT sufficient

Need at least 2048×2048.

10.7 Colour Chromaticity Calculation — Dec 2024**RGB colour for given chromaticity (x, y, z = 1-x-y)**

Warm white (0.45, 0.40): $z = 0.15 \rightarrow R=45\%, G=40\%, B=15\%$

Deep blue (0.15, 0.20): $z = 0.65 \rightarrow R=15\%, G=20\%, B=65\%$

10.8 Pixel Resolution Calculation — Dec 2024**Smartphone: 1080×1920 px, 5-inch diagonal**

Diagonal px = $\sqrt{1080^2 + 1920^2} = \sqrt{4,852,800} \approx 2203$ px

PPI = $2203 / 5 \approx 441$ PPI

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UNIT 11: Exam Strategy & Most Likely Questions

11.1 Paper Pattern

Question	Pattern
Q1 (Compulsory)	5 sub-parts of 8 marks each (or 4-6 parts of varying marks) = 40 marks
Q2 – Q5 (Attempt any 3)	3 sub-parts per question; 20 marks each → 60 marks
Total	100 marks, 3 hours, Scientific calculator allowed

11.2 Most Probable Questions for Next Exam

Based on 7-semester frequency analysis, prepare these topics thoroughly:

Probability	Expected Question
★★★ Certain	Image digitization with sampling & quantization + numerical (aspect ratio / DPI)
★★★ Certain	DFT: properties + numerical computation (1-D or 2-D)
★★★ Certain	Image enhancement: histogram equalization + spatial filters (Laplacian / Gaussian)
★★★ Certain	Noise models: at least Gaussian, Salt & Pepper, one more — with PDF formulas
★★★ Certain	K-Means clustering with step-by-step numerical example
★★★ Certain	Supervised vs Unsupervised learning — comparison table + categories
★★★ Certain	Colour models: RGB + HSI + CMY — at least one comparison + formula
★★ Very Likely	Wavelet Transform: CWT vs DWT, properties, comparison with Fourier
★★ Very Likely	Frequency domain filters: LPF, HPF, GLPF, Butterworth — comparison
★★ Very Likely	Agglomerative hierarchical clustering with dendrogram, single/complete link
★★ Very Likely	Camera calibration / pin-hole model / single vs stereo camera
★★ Very Likely	Edge detection: Canny, Sobel, comparison of algorithms
★ Likely	Wiener filter — formula and problem it solves
★ Likely	Homography — definition + applications

★ Likely	DCT matrix of order 2 + advantages over DFT
★ Likely	Feature extraction — types and role in object detection

11.3 Short Notes Rotation (Q5 — pick 5 of 6)

Every paper has a short notes question. Prepare ALL of these as any 5 may appear:

- Gaussian noise (5 papers)
- K-Means clustering (4 papers)
- Median filter / Mean filter (4 papers)
- Gaussian LPF / Butterworth LPF / Ideal LPF (4 papers)
- Supervised learning / Bayesian classification (4 papers)
- Colour models — HSI, RGB, CMY, pseudo colour (3 papers)
- Pixel resolution / Spatial resolution / Intensity resolution (3 papers)
- Salt and Pepper noise / Rayleigh noise (3 papers)
- Edge-based segmentation / Region detection (3 papers)
- Haar transform / DCT (3 papers)
- Stereovision / Camera calibration (2 papers)
- Homography / Feature extraction / Linear discriminant function (2 papers)

APPENDIX: Quick Formula Reference Card

Transforms

DFT: $F(u) = \sum f(x) e^{-j2\pi ux/N}$ IDFT: $f(x) = (1/N) \sum F(u) e^{j2\pi ux/N}$
 2D DFT: $F(u,v) = \sum \sum f(x,y) e^{-j2\pi(ux/M+vy/N)}$
 DCT: $C(u) = \alpha(u) \sum f(x) \cos[(2x+1)u\pi/2N]$
 Haar N=2: $H = (1/\sqrt{2}) \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
 Wavelet: $\psi_{\{a,b\}}(t) = (1/\sqrt{a}) \psi((t-b)/a)$

Enhancement

Histogram Eq: $s_k = (L-1) \sum p(r_j) \quad j=0..k$
 Contrast stretch: $s = (r-r_{\min})(L-1)/(r_{\max}-r_{\min})$
 Laplacian: $\nabla^2 f = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y)$
 Sharpen: $g = f - \nabla^2 f$ (negative centre mask)

Noise & Restoration

Gaussian PDF: $p(z) = (1/\sqrt{2\pi\sigma^2}) e^{-(z-\mu)^2/2\sigma^2}$
 Degradation: $G(u,v) = H(u,v)F(u,v) + N(u,v)$
 Inverse filter: $\hat{F} = G/H$
 Wiener filter: $\hat{H}_w = H^* / (|H|^2 + K) \quad K = S_n/S_f$

Colour

RGB→CMY: $C=1-R, M=1-G, Y=1-B$
 RGB→HSI: $I=(R+G+B)/3, S=1-3 \cdot \min(R,G,B)/(R+G+B)$
 Chromaticity: $x+y+z=1$

Clustering & Classification

K-Means objective: $J = \sum_j \sum_i \epsilon_j ||x_i - \mu_j||^2$
 Euclidean: $d = \sqrt{\sum (x_i - y_i)^2}$ Manhattan: $d = \sum |x_i - y_i|$
 Bayes: $P(\omega_j|x) = P(x|\omega_j)P(\omega_j)/P(x)$
 LDA: $g(x) = w^T x + w_0 \rightarrow \text{class } \omega_1 \text{ if } g(x) > 0$

Camera & Vision

Pinhole: $u = fX/Z$, $v = fY/Z$
Disparity: $d = x_L - x_R$ Depth: $Z = fB/d$
Homography: $x' = Hx$ (3×3 matrix, 8 DOF)

END OF BOOK — Good luck in your examination!

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